

Deep Learning-Based Brain Tumor Analysis: Segmentation and Segmentation-Guided Classification of Tumors

A PROJECT REPORT

Submitted by

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ABSTRACT

This work presents a multi-task deep learning model on the automated recognition and classification of brain tumors in MRI scans. Manual diagnosis of brain tumors is usually time consuming, unreliable and very careful to the interpretation by experts, particularly, because of the different shapes, sizes, and appearances of tumors. To address these shortcomings, the model will combine the binary segmentation with multi-class classification in one system. It is built on the foundations of an improved U-Net, with attention and residual connectedness and squeeze-excitation modules to enhance feature identification and concentration on areas of interest containing tumor. The segmentation head produces a binary mask which is accurate in localizing the tumor area and the classification head is a Global Average Pooling layer, then dense layers followed by a softmax output that classifies the tumor as Glioma, Meningioma, Pituitary, or No Tumor. It is trained on a selected MRI dataset comprising 6,380 balanced samples and is tested with the help of such metrics as Dice coefficient, IoU, and classification accuracy. As the experimental results indicate, the proposed hybrid model provides better performance than the conventional U-Net variants, meaning that it is able to combine spatial accuracy with discrimination at the class level. This is not only an accurate diagnostic method but also a powerful, effective, and scalable method to support radiologists in making clinical decisions.

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Chapter 1

INTRODUCTION

Brain tumors constitute one of the most significant medical issues in neuroimaging and clinical oncology as they are complex in structure, irregularly defined, and diversified in terms of tissue properties. The correct identification, location and classification of these tumors are essential towards proper treatment planning and prognosis. Manual interpretation of MRI (Magnetic Resonance Imaging) scans by radiologists is however usually time-consuming, tedious and also subjected to human biases especially where the tumor has similar visual features with normal tissue. The above challenges have prompted the necessity to have an automated, objective and very precise system that can aid medical professionals to diagnose brain tumors effectively.

The application of deep learning in medical image analysis has already transformed the sphere of computer-aided diagnosis systems to reach the level of professional medical accuracy. Convolutional Neural Networks (CNNs) are some of them, as well as encoder-decoder models such as U-Net, have demonstrated outstanding results in medical segmentation and classification tasks. Although there has been these improvements, the traditional models do not distribute tumor regions and classifications of tumor types in one framework. The constraint is the impetus to design a multi-task deep learning network that can do both pixel-wise segmentation and image-level classification simultaneously. Through the combination of these two tasks, the model will be able to use shared features, enhance generalization, and minimize computational redundancy.

To deal with these issues in this project, the U-Net architecture has been improved by adding residual connections, attention gates, and squeeze-excitation modules. The segmentation arm produces a binary mask identifying the tumor and non-tumor regions and classification arm produces global feature representation identifying the tumor as Glioma, Meningioma, Pituitary or No Tumor. With the attention mechanism, the network is capable of highlighting the most informative parts of the image, and the residual and squeeze-excitation blocks reinforce the propagation of features and eliminate unwanted noise. The dataset, aggregated based on various open-access MRI repositories has been thoroughly preprocessed, balanced and standardized to provide uniformity and strength when comparing training and testing.

Generally, the proposed model is a coherent and effective way of analyzing brain tumors through integrating binary segmentation with four-class analysis in a single pipeline. It saves the necessity of different models and simplifies the workflow of the diagnostics, thus providing more rapid and precise

solution to the clinical decision-making. Having better Dice scores, IoU, and classification accuracy, the system proves to provide good predictions and segmentation mask of good quality. This solution is a major leap towards the real-world use of AI-based radiological technologies that can improve diagnostic accuracy and offer real-time assistance to medical workers.

1.1 Motivation

The rationale of this work is the fact that the diagnosis of brain tumours is often limited by the shortcomings of manual analysis of MRI images, that are based on the urgency, accuracy, and reliability of such diagnoses. The problem of brain tumors is life threatening and therefore necessitates the detection and clear classification at an early stage in order to implement effective treatment measures. Nevertheless, the radiologists have a lot of difficulties in reliably visualizing tumors because they have irregular shapes, a range of intensities and visual resemblance to normal brain tissues. This project solves these problems by using deep learning to also automate tumor segmentation and classification so that manual interpretation is no longer necessary. Through the addition of attention mechanisms, the residual learning and squeeze-excitation modules to a multi-task U-Net architecture, the suggested system is capable of improving the accuracy of the diagnostic result as well as reducing human error, speeding up the analysis process, and offering an effective clinical decision-support tool to be used in a real-world medical setting. This work is significant as it can make medical imaging a more accurate, efficient, and affordable way of making a diagnosis and eventually enhance the results of patients by helping to diagnose them early and correctly.

1.2 Objectives

- To create a deep learning-based automated system that can identify and classify brain tumors using MRI scans with high precision, reducing the amount of manual work and diagnostic time.
- To come up with a binary segmentation model that accurately locates and marks tumor areas in MRI scans, it is important to allow correct localization of abnormal tissues.
- To apply a 4-classification model of MRI images distinguishing between Glioma, Meningioma, Pituitary or No Tumor to allow complete determination of the type of tumor.
- To combine the segmentation and classification tasks in a single multi-task U-Net model that is more computationally efficient and better feature sharing between the tasks.
- To enhance the conventional U-Net structure with the help of the attention gates, the residual connections, and squeeze-excitation modules to learn the features of the relevant tumor with a more focused and efficient manner.
- To preprocess and balance MRI data taken in multiple reputable sources, it is important to ensure consistency in image size, the intensity normalization, and the quality of annotation.

- To train and test the model on strong measures like Dice coefficient, IoU, precision, recall, and classification accuracy to measure the overall performance across the board.
- To visualize model predictions by bringing segmentation masks onto MRI images, to have interpretable results on which to perform a qualitative analysis and validation.
- To develop an end-to-end system that provides clinically useful predictions, to provide a support tool to the radiologist in both diagnosis and treatment planning.
- To increase the speed of the diagnostic process and minimize the subjectivity of the human evaluation, it is crucial to encourage the consistency between various cases and institutions.
- To investigate the model scalability to realize the model as a real-time clinical decision-support system, which may be later applied to hospitals and telemedicine platform.

Chapter 2

LITERATURE SURVEY

The literature review is based on recent developments in brain tumor detection and segmentation with the help of deep learning and hybrid machine learning models applied to MRI data sets. It describes how past approaches to machine learning were developed to the current state-of-the-art CNN, U-Net, transformer, and GAN-based models which have a markedly better diagnostic specificity, segmentation accuracy and computational efficiency. All these studies set an importance on integrating hybrid frameworks, attention mechanisms and explainability techniques to promote the clinical reliability and interpretability. The review also presents the current challenges to be dataset imbalance, low contrast, and model generalization and provides emerging trends such as multimodal fusion, domain adaptation, and federated learning to generate robust and scalable analysis of brain tumors.

2.1 Literature Review

Mona et al. [1] introduced the latestnnU-Net that uses residual blocks, attention gates, and Hausdorff distance loss to achieve accurate segmentation of brain tumors. The use of LIME-based explainability also made brain tumorcores of 0.83 and 0.71, respectively, and it outperformed the baseline models. The method improved localization at the boundaries and extraction of spatial features which were successful in adult and pediatric MRI segmentation. Mohammad et al. [2] proposed the VGG-SCNet, a hybrid deep learning system that includes VGG-16 with ensemble machine learning models, such as AdaBoost and XGBoost, to detect brain tumors through the MRI. This model was 99.2% precise and 99.1% recall, which was better than baselines of transfer learning. It exhibited high levels of robustness and thus it was the one that could be used in automation of clinical diagnosis. Kittipol et al. [3] designed U-Net++DSM, which is an improved architecture that incorporates Deep Supervision Mechanism (DSM) and dilation operators to achieve a precise tumor segmentation. On BraTS and Kaggle datasets, the model scored 98.64 and 98.02% accuracy and Dice, respectively. Its steps of denoising and normalization enhanced accuracy but reduced overfitting. Anindya et al. [4] suggested TumorGANet as a hybrid network capable of extracting features with ResNet50 and data augmentation and transfer learning using GAN. It attained an accuracy of 99.53, precision of 100 percent and recall and a Hamming loss of 0.2 percent. The use of LIME-based explainability also made brain tumor analysis of clinical data more transparent and interpretable.

Ruqsar et al. [5] suggested a deep learning-based model RU-Net2+ that incorporates Convolutional Normalized Mean Filtering (CNMF), DBT-CNN and Cox-logistic regression to provide precise brain tumor segmentation and survival. The model performed very well, classified at 99.51% (HGG) and 99.28% (LGG) with survival predictions on the long-, medium-, and short-term cases of 85.71, 72.72 and 61.54 respectively. It similarly enhanced both the accuracy of segmentation and the accuracy of prognosis, which is more predictive than the current approaches. Metlek et al. [6] have proposed ResUNet+, a hybrid attention-based versus convolutional network that can segment brain tumors using the BraTS 2018 / 2020 datasets. The model has residual and node blocks that have ROI-based preprocessing to enhance extraction of features and also minimize the computation cost. It scored 92.80% in Dice on (WT), 93.10% in Dice on (ET) and 91.90% on (TC), superior to standard UNet models. The improved structure highly reduced the gradient problems and segmentation strength. Nashaat et al. [7] created an effective hybrid system of the fuzzy logic and deep learning that detects and classifies brain tumors using MRI scans. Their method combines fuzzy thresholding based on precise segmentation and an optimized CNN based on classification of glioma, meningioma and pituitary tumors. The model had a 98 percent accuracy and a 97.97 percent Dice coefficient, which is better than the traditional ML and DL models. Joice et al. [8] conducted an in-depth review of machine learning and deep learning algorithms in brain tumor classification based on MRI images. The authors compared several architectures such as CNN, ResNet, DenseNet, EfficientNet, YOLO, and Vision Transformers to the BraTS dataset. Accuracies up to 98.6% were obtained with CNN-based methods and up to 99.2 with transformer-based methods, focusing on the trade-off between the efficiency of the computation and diagnostic quality. The paper also addressed the issues of clinical deployment and the research directions in AI.

Mallampati et al. [9] created a hybrid ML model comprising of 3D-UNet segmentation requirements along with KNN and Gradient Boosting Classifier to detect brain tumors. The model made use of the shape, texture as well as statistical MRI properties in order to improve precision. It obtained 71% accuracy of classification, which was better than the 2D-based methods. This hybrid structure enhanced quality and strength to medical diagnosis. Pratikkumar et al. [10] have provided an extensive survey of the use of the ML, CNN, and transformer-based models in brain tumor classification via MRI. The review analyzed more than 60 studies (2018-2023) and found the best accuracies of 99.3 on BraTS and AANLIB datasets. It highlighted the lack of generalization and interpretability. The paper pointed out hybrid deep models as the prospective path. Zheshu et al. [11] suggested a framework of FAHS-SVM that combines morphological filtering, wavelet transform, and Extreme Learning Machine to classify brain tumors. It had an accuracy of 98.51% to distinguish normal and abnormal tissues. The method minimized misclassification and enhanced uniformity of segmentation. It was better at the precision and speed than CNN and DNN. Ayesha et al. [12] made an extensive survey of deep learning methods of brain tumor classification based on the MRI files. The article has analyzed CNN, GAN, and hybrid networks and provided the summary of findings in 50+ (2018-2023) studies. Accuracies were reported to be 99.7% on models such as ResNet and BrainMRNet. It concluded that DL models are superior to the traditional ML, and clinical standardization is advisable.

Zhihua et al. [13] provide a survey of deep learning-based brain tumor segmentation that evaluated

more than 150 studies focusing on such architectures as U-Net, DeepMedic, and V-Net. Low contrast, annotation bias, and imbalanced data were also highlighted as challenges in the paper, and Dice scores of over 0.9 on BraTS datasets were also noted. It proposed domain adaptation and search over neural architecture as important future directions towards better generalization. Kim et al. [14] suggested TACA-RNet, which is a Tri-Axis-based Context-Aware Reverse Network that combines axial, coronal, and sagittal MRI planes to segment multimodally. It increased the contextual and boundary accuracy with modules such as TACR, MSCF, and ARA. The model was also successful on BraTS 2018/2020 with Dice scores of 84.06% and 83.49%. Yahya et al. [15] studied the 169 methods of segmentation of brain tumors, divided them into manual, semi-automatic, and deep learning-based approaches. CNN and hybrid architectures including U-Net and FCN were the most useful as the review found out. It concluded that deep learning together with hybrid frameworks showed greater segmentation accuracy across MRI modalities. Abidin et al. [16] conducted a review of CNN, transformer and hybrid brain tumor segmentation with multi-modal MRI. Analyzing the latest 2021/2023 frameworks, the study showed that the hybrid CNNViT architecture had a dice score of more than 0.9. It has drawn attention to such matters as an imbalance in datasets and their inability to be interpreted, calling on the need to pay attention to explainable and adaptive models.

Magadza et al. [17] observed the developments in the field of deep learning architectures to brain tumor segmentation, including CNN, FCN, and U-Net variants. It investigated dual-path, cascaded, and ensemble, where it reported Dice scores greater than 0.9 on BraTS datasets. The paper highlighted the superiority of U-Net-based ensembles and the necessity to address annotation and computational challenges to use it in clinical practice. Dorfner et al. [18] conducted a review of deep learning applications in the analysis of brain tumor including segmentation, classification, and biomarkers prediction based on MRI data. The paper identified the transition between radiomics-based ML and CNN and transformer models. It covered such issues as the imbalance of data and explainability, and it suggested foundation models to enhance reliability. The authors made a conclusion that the use of DL allows non-invasive and individual neuro-oncology diagnostics. Bharathi et al. [19] performed a survey of deep learning-based brain tumor segmentation, paying attention to multi-modal MRI to improve diagnosis. It presented a 3D multi-modal segmentation network based on CNN, RNN, and GAN to provide strong classification. The model enhanced the efficiency of diagnostic and minimized time of computation. It showed the possibilities of semi-supervised learning in medical images.

Elaissaoui et al. [20] conducted a survey of the deep learning architectures (such as CNN, U-Net and Capsule Networks) in brain tumor detection with MRI images. Dice scores were reported to be 0.93 with classification accuracies of 99 percent using ResNet and IACO-ResNet. The paper had observed difficulties with the lack of data and interpretability. It promoted the use of cloud fog in real-time analysis of medical images. Mohsen et al. [21] conducted a review of AI-based brain tumor detection based on MRI, CT, and PET in terms of CNN, RNN, GAN, and transformer frameworks. This paper covered such datasets as BraTS and TCIA and their diagnostic functions. Heterogeneity and clinical deployment were some of the major challenges. It found the future research trends as federated learning and multimodal fusion. Rasool et al. [22] provided a state-of-the-art review on the deep learning-based brain tumor classification (2020/2024) with the MRI datasets. Such methods

as GoogleNet, ResNet50 or MCNN reached accuracy rates of 99.89. The paper identified CNN as being more superior in automatic feature extraction. It has observed that different datasets and 3D implementations were necessary. Sabaa et al. [23] conducted a systematic review of CNN-based MRI brain tumor techniques and examined 427 articles in large databases. Neural networks such as U-Net, ResNet and DeepMedic were able to score up to 0.96 on BraTS datasets. The paper affirmed that CNN was more accurate and efficient. It suggested performance improvement with transfer learning and data augmentation.

2.2 Research Gap

Although outstanding progress has been made in the field of deep learning in detecting and segmentation of brain tumors, there are still some research gaps. Majority of the models require the use of datasets such as BraTS which is not diverse in terms of scanners, modalities and demographics resulting in low generalization on clinical data. Hand labeling is also tedious and inaccurate, leading to annotation bias. There are also additional challenges in deep models of low-contrast areas, abnormal tumor shape, and tissue boundaries. Most frameworks are not scalable since they need big datasets and high computing abilities. The edge localization and the small tumors continue to be a problem. Therefore, better generalization, diversity of the dataset, and embedded segmentation are all critical aspects to be developed in the future.

The interpretability and transparency of AI-based diagnostic systems is another big gap. Most deep learning models are black box, which reduces clinical trust and acceptance, despite the high levels of accuracy they provide. Not many of them use explainable AI methods to substantiate their predictions. Real time flexibility and computational efficiency also is an issue with the application. Multimodal data including MRI, CT and PET are not well integrated. Federated learning methods that preserve privacy are not well studied with respect to medical cooperation. It is essential to deal with these shortcomings using explainable, adaptive, and hybrid frameworks to achieve applications in the real world.

Chapter 3

SYSTEM REQUIREMENTS

To implement and train the proposed multi-task deep learning model, the system needs to have a contemporary computing environment with GPU acceleration support. It should run on a 64-bit operating system like Windows 10/11, Ubuntu Linux, or MacOS and should have Python (3.8 or later) and other principal deep learning packages, such as TensorFlow/Keras or PyTorch, OpenCV, NumPy, and Matplotlib to process and visualize images. One can work and experiment with additional tools like the Jupyter Notebook and an IDE like VS Code or PyCharm. The storage capacity of the MRI datasets and preprocessed images, model checkpoints, and experiment logs require a minimum of 20-30 GB of free space. It is also important to have CUDA and cuDNN (in the case of NVIDIA GPUs) properly installed so as to be able to train large image datasets efficiently.

On the hardware side, a dedicated GPU can be of much good when training the improved U-Net with attention, residual as well as squeeze-excitation modules. An 8 GB or more VRAM GPU (like NVIDIA GTX/RTX models) with 16GB or more RAM on the system and a multi-core processor is suggested to support the loading of data, preprocessing and model training without issues. To do simple inference or for use as a demo (e.g. in a hospital setting or via a web interface), the model can be executed on a less powerful machine or even CPU-only, but with higher prediction times. This could require a stable internet connection in case the system is built into a cloud implementation or a web-based clinical decision-support system. These system requirements will, in general, guarantee that the model may be trained effectively, assessed with high reliability and deployed in a manner that would permit real-time or close to real time analysis of brain tumors.

Chapter 4

DATASET DESCRIPTION

This chapter presents the dataset used for training, validating, and evaluating the proposed multi-task deep learning model for brain tumor analysis. The model is designed to perform two primary tasks: binary semantic segmentation of tumors and four-class classification of tumor type. To enable this, the dataset must contain both pixel-level mask annotations and image-level class labels. The dataset selected for this project is a publicly available MRI brain tumor dataset that meets these requirements. It contains well-preprocessed, class-balanced images distributed across multiple anatomical orientations, making it an ideal foundation for developing a generalized and reliable computer-aided diagnosis system.

4.1 Mixed Brain Tumor Dataset

The Mixed Brain Tumor Dataset comprises a total of 6,380 T1-weighted contrast-enhanced MRI scans. Each image in the dataset is paired with a corresponding binary segmentation mask and a tumor type classification label. The dataset is evenly split into four distinct tumor classes—No Tumor, Glioma, Meningioma, and Pituitary—with exactly 1,595 images in each category. This class balance eliminates the risk of model bias during training and ensures that the classifier performs uniformly across all categories.

A unique strength of this dataset lies in its diversity of anatomical perspectives. MRI scans are presented in three primary orientations: axial (cross-sectional view from top to bottom), sagittal (side view from left to right), and coronal (front-to-back view). These orientations are uniformly represented across all tumor classes, which enhances the model’s ability to recognize tumor structures from multiple anatomical angles. All images have been resized to a consistent resolution ($160 \times 160 \times 3$) and normalized for intensity, ensuring uniformity during training.

4.1.1 No Tumor Class

This category includes brain MRI scans that do not exhibit any signs of tumor growth. These images are critical for helping the model distinguish between healthy and pathological brain structures. The segmentation mask for each image in this class is empty (all pixel values are zero), reflecting

the absence of abnormal tissue. These non-tumor scans are particularly valuable in minimizing false positives during model prediction. Each image is captured in either axial, sagittal, or coronal orientation, maintaining a balanced view distribution to assist the model in learning what constitutes a healthy brain from all standard clinical perspectives.

4.1.2 Glioma Class

The Glioma class features MRI scans from patients diagnosed with glioma, a common and often malignant tumor type originating from glial cells. Gliomas can present with irregular and diffuse boundaries, varying in shape and intensity, which makes their segmentation particularly challenging. In this class, each MRI image includes a corresponding binary mask that delineates the tumor region. Additionally, each image is labeled with the classification tag “Glioma.” The dataset ensures balanced inclusion of axial, sagittal, and coronal views, helping the model to learn the varied morphology of gliomas across multiple planes of visualization.

4.1.3 Meningioma Class

Meningiomas arise from the meninges—the protective membranes that cover the brain and spinal cord. These tumors are typically well-encapsulated and benign, but their size and proximity to critical brain regions can still result in serious health issues. The MRI scans in this class include binary masks marking the tumor boundaries, which are often more distinct compared to gliomas. Each image is labeled with the “Meningioma” class tag and is equally distributed across the three anatomical orientations. This enables the model to generalize the appearance of meningiomas regardless of the imaging angle, enhancing robustness in diverse clinical scenarios.

4.1.4 Pituitary Class

The Pituitary class includes MRI scans that exhibit tumors located in or around the pituitary gland. These tumors are generally smaller in size but are clinically significant due to their location near vital hormonal control centers such as the hypothalamus and optic chiasm. Accurate detection of pituitary tumors is critical, as even small growths can disrupt endocrine function. Each image in this category is annotated with a binary mask that highlights the tumor region and is labeled with the “Pituitary” classification. The dataset provides a uniform distribution of pituitary tumor images across axial, sagittal, and coronal orientations, ensuring that the model learns to detect these tumors even when they are viewed from challenging angles.

This comprehensive and balanced dataset provides a strong foundation for developing a deep learning model that is capable of not only accurately localizing tumors via segmentation but also correctly identifying their type through classification. The diversity in orientation and uniform class distribution make this dataset highly suitable for building generalized models with strong clinical applicability.

Chapter 5

CENTRALIZED DEEP LEARNING SYSTEM SETUP

In this chapter, the design and training pipeline of the proposed centralized deep learning system developed on brain tumor segmentation and classification with the MRI scans are elaborated on. The system takes advantage of multi-task learning through a single computational environment to effectively and efficiently learn two tasks: (i) binary semantic segmentation to locate tumor regions and (ii) four-class classification to recognise tumor type (Glioma, Meningioma, Pituitary or No Tumor). Having all the workflow in one platform is important to maintain consistency in data manipulation, simplified experimentation, and reproducibility of the results in every arrangement.

It is also modular and extensible, and thus can be used to evaluate various types of backbones in the same pipeline: the baseline U-Net, Residual U-Net, and Attention U-Net. The figure 5.1 shows the general system design that consists of data loading, preprocessing, shared encoding, task-specific decoding, loss calculation, and optimization of the model in an iterative manner. The system supports end-to-end training and evaluation, starting at the stage of raw image ingestion and concluding with the reporting of its performance.

5.1 System Pipeline and Workflow

This model starts by loading MRI images, segmentation masks, and class labels of tumors in local storage. All the images are preprocessed, which involves image scaling to 160 x 160 pixel, equalization of image intensity, and conversion of masks to binary format. Also, each image takes a specific label as a tumor. The MRI images are shot at three clinical orientations such as axial, sagittal, and coronal that are equally distributed among the four classes that offer the model a great and diverse anatomical representation.

After preprocessing, data augmentation methods in options like rotation, flipping, and zooming are used to enhance the robustness and avoid overfitting. The standardized and augmented images are processed by a common encoder which identifies both the spatial and contextual features extracted to aid segmentation and classification activities. The result of the encoder is then inputted to two parallel lines:

- A pixel-level binary tumor mask prediction segmentation decoder.
- A classification head which uses global pooling and fully connected layers to forecast the type of tumor.

The process of training is optimistic in that the two tasks are optimized together by minimizing a joint loss. Segmentation is performed with the help of Binary Cross-Entropy (BCE) and Dice loss, whereas classification is performed with the help of Categorical Cross-Entropy (CCE). All these losses are added or weighted to formulate the end loss to be used during backpropagation

5.2 Training Workflow Overview

Figure 5.1 demonstrates the entire training process and identifies every step of the process, starting with the initialization and concluding with the convergence:

- **Start System:** The start System initializes all the necessary components such as data paths, model architecture and optimizer.
- **Load MRI Scans, masks, labels:** Import MRI raw data and separate it into training, validation and testing data.
- **Preprocessing:** Preprocess the input data, e.g., by resizing images, performing intensity normalization, and converting masks to binary and class labels to codes.
- **Data Augmentation:** Use augmentation methods to augment data and eliminate overfitting.
- **Shared Encoder (Feature Extraction):** Deeply spatial features of images are extracted to be shared between the two decoders.
- **Segmentation Decoder:** produce a binary mask, which marks the tumor region.
- **Classification Head:** Given the extracted features, predict one among four tumor classes.
- **Loss Computation:** Calculate segmentation loss (BCE + Dice) and classification loss (CCE), and sum them up.
- **Backpropagation:** Calculate gradient descent and update the weights of models.
- **Checkpointing:** Store the model weights as validation performance is increasing.
- **Convergence Check** Check whether the performance measures have become stabilized.
- **End Training:** Store the trained model at convergence.

5.3 Flowchart Explanation

Figure 5.1 below is the visual representation of the training and system workflow. The diagram illustrates the overall process followed to the end when training is being done and it gives importance to the preprocessing and architectural elements.

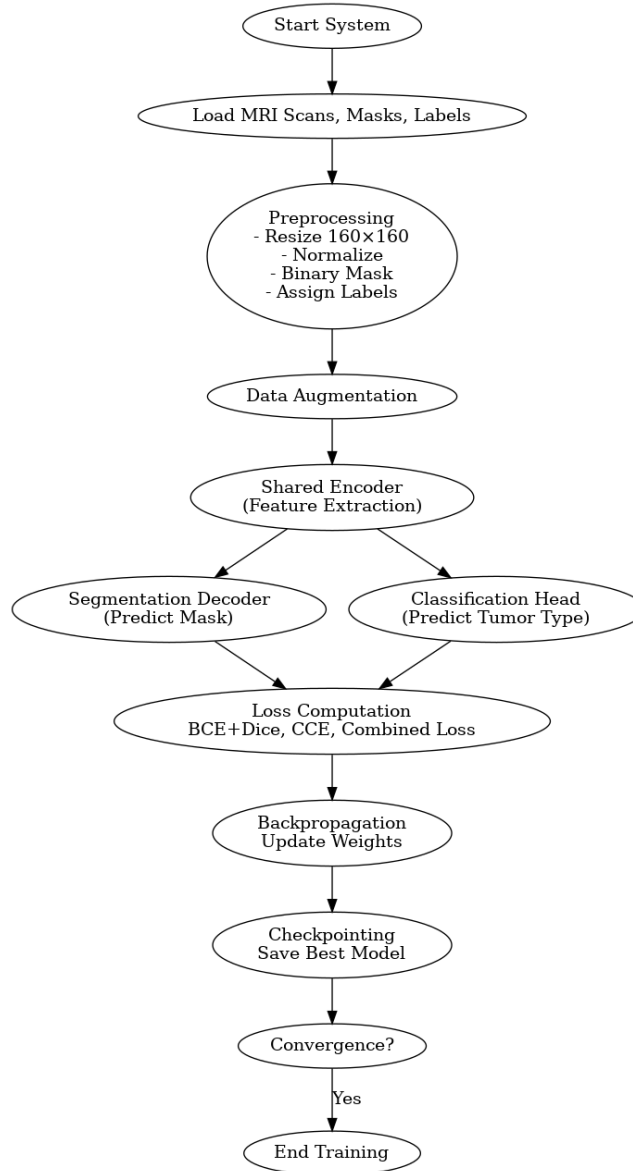


Figure 5.1: System Training Flowchart

The flowchart is made up of the following main elements:

- Start System: Startup of runtime environment and system components.
- Load MRI Scans, Masks, Labels: Data consumption into memory out of organized directories.
- Preprocessing: The images are resized to 160x160, normalized by the intensity and combined with binary masks and classification labels.
- Data Augmentation: Pictures are enhanced to enhance generalization.

- Shared Encoder: Generates low to high level properties of the MRI input.
- Segmentation Decoder: Produces binary tumor mask with encoder features.
- Classification Head:/ Specifies tumor class (Glioma, Meningioma, Pituitary, or No Tumor).
- Computation of loss- loss: This is the sum of segmentation and classification loss.
- Backpropagation: The model weights are updated with the calculated gradients.
- Checkpointing: Stores the optimum model weights via validation.
- Convergence Check: Training is stopped when the model performance is stable.
- End Training: Finalizes the process and returns the trained multi task model.

This is a structured and centralized system design that guarantees the learning pipeline is efficient, scalable, and reproducible, which are necessary qualities of clinical use of brain tumor analysis. The system is able to combine segmentation and classification into a common system that is used to optimize the value captured per MRI scan and hence more informative and accurate predictions are made.

Chapter 6

PROPOSED ARCHITECTURE

6.1 Overview of the Proposed System

The proposed system will allow to cope with the issue of binary semantic segmentation of brain MRI images, but the image-level classification will also be conducted to allow contextual awareness. Through the multi-task learning approach, the system incorporates two output heads, one of which is used to perform pixel-wise tumor segmentation and the other to perform binary classification (tumor vs. no tumor). Such two-headed design allows the model to learn not only local spatial features but global contextual information that enhances the precision of segmentation, boundary demarcation as well as overall strength.

The architecture is based on a convolutional encoder-decoder architecture based on U-Net that is appropriate in biomedical image segmentation as it has a symmetrical design and skip connections. The encoder reduces spatial resolution but retrieves semantic features and the decoder restores fine-grained segmentation masks by successively restoring spatial detail. The encoder bottleneck is filled with the classification head, which uses the most abstracted features to predict the whole image. This similar representation in both tasks gives the model the ability to yield a more generalized representation when different tumors appear, location, and orientation.

Three variants of U-Net that were customized were implemented and trained on a unified training pipeline to assess the results of architectural changes: (i) Base U-Net, (ii) Residual U-Net, and (iii) Attention Residual U-Net. Both the models preserve the dual-head structure and they are end-to-end trained using combined loss functions.

6.2 System Architecture Overview

Figure 6.1 (end-to-end architecture of the proposed system) is a visual summary of the system. It shows all steps involved in the workflow such as ingestion of raw data to clinical decision support, and the way the various modules are entangled to enable dual-task learning in brain tumor analysis. The architecture is separated into five major blocks which are: Data Source, Data Preprocessing, Training Pipeline, Inference Pipeline, and Evaluation Module.

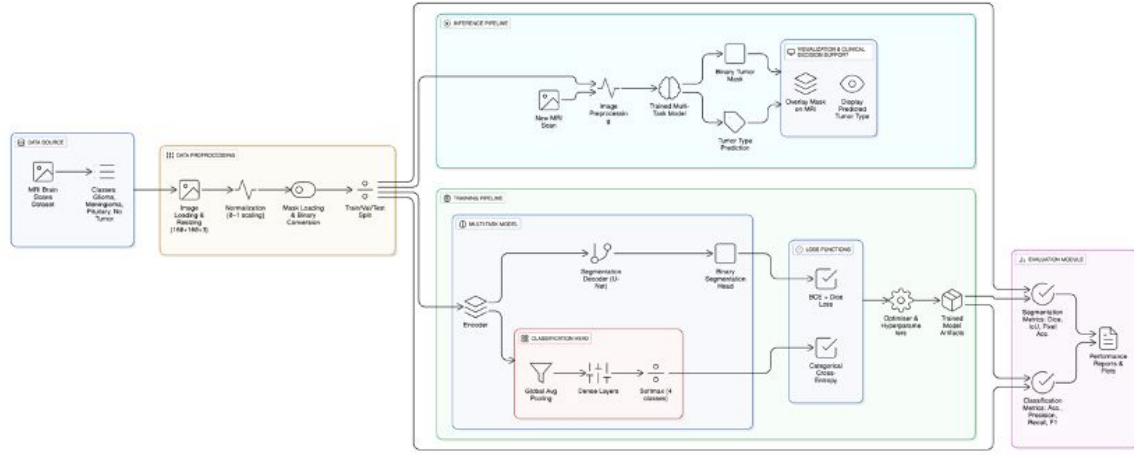


Figure 6.1: System Architecture

- **Data Source:** The system starts with an organized dataset of brain MRI scan that has four balanced classes including Glioma, Meningioma, Pituitary and No Tumor. The images have a segmentation mask and a classification label in each image annotation and the dataset is captured at axial, coronal and sagittal orientation to enable the generalization of the model.
- **Data Preprocessing:** MRI-scans are resized to a standard size of 160x 160x 3. Intensity normalization converts pixel values to the range [0, 1], making the samples have constant contrast. Segmentation masks are binarized to distinguish between tumor and background and the data is divided into training, validation, and test subsets. This initial processing is done to make sure that the input data meets the requirements of the model and improves convergence of the training.
- **Training Pipeline:** Its main training aspect is a multi-task model based on the form of U-Net (Base, Residual, or Attention Residual). The input undergoes a common encoder which generates hierarchical features. These characteristics are transmitted by two parallel output heads:
 - The Segmentation Head is a reconstruction of pixel-level tumor masks based on transposed convolutions and skip connections.
 - Classification Head entails global average pooling and then dense layers to determine the type of tumor with the help of a softmax activation function.

Each task is computed separately with a combination of Binary Cross-Entropy (BCE) and Dice Loss in segmentation, and Categorical Cross-Entropy in classification. Such losses are propagated backward together to update the model weights with an Adam optimizer with hyperparameters optimized by hand. The model checkpoints are stored in the best performance.

- **Inference Pipeline:** To be used in practice, the system can be scaled to new MRI scans inference. Images that come in are first processed in the same manner as a training image and fed through

a trained multi-task model. It has the outputs of binary tumor mask and predicted tumor class. These are then visualized i.e. by superimposing the predicted mask on the MRI image and showing the predicted class label hence help in clinical decision-making.

- **Evaluation Module:** The system is strictly tested in terms of segmentation and classification statistics. In cases of segmentation, they are reporting Dice Score, Intersection over Union (IoU), and pixel accuracy. To measure the performance of the model, accuracy, precision, recall, and F1-score are to be used. The entire results of evaluation are summarized into detailed plots and performance overviews to aid in the analysis and comparison of variants of the model.

This is a system level perspective that shows the modularity, scalability and real applicability of the suggested architecture. Multi-task learning is well-placed to work with clinical workflows and future studies on automated brain tumor analysis because of the combination of a constant preprocessing and a healthy evaluation.

6.3 Custom U-Net Variants

- **Base U-Net:** The backbone is based on the original U-Net design, and has a symmetric encoder-decoder architecture with skip connections between matching encoder and decoder blocks. The encoder comprises of convolutional layers, which are then followed by max-pooling in order to downsample the input and obtain features, and the decoder adheres to the same pattern with upsampling and convolution aimed at reconstructing the output mask. The bottleneck is further added with the help of global average pooling and fully connected layers to generate a binary result. It is a basic version that can be used as a baseline to evaluate performance when performing a segmentation task with no extra complexity.

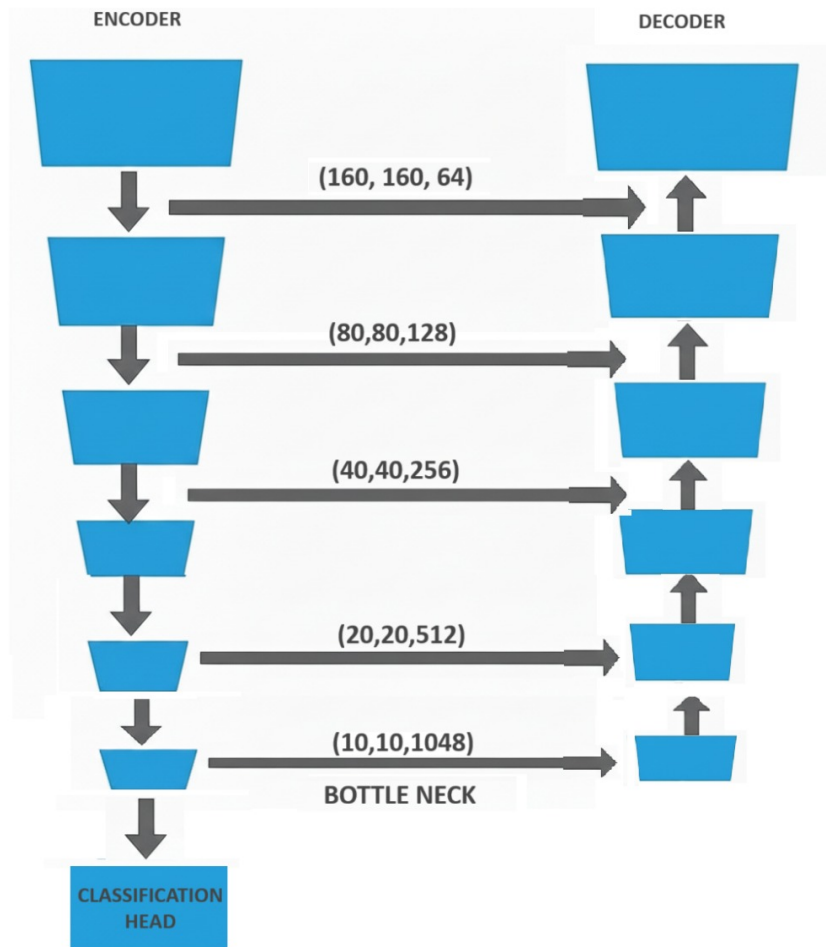


Figure 6.2: Architecture of Base U-Net

- **Residual U-Net (ResU-Net):** The Residual U-Net is an extension of the base model with residual blocks instead of the regular convolutional blocks and identity shortcut connections. These residual pathways permit gradients to more easily flow during backpropagation and eliminate the vanishing gradient problem, making it possible to learn more about features. Segmentation head has more valuable intermediate representations and classification head receives more discriminative bottleneck features. The architecture enhances the convergence of the model, segmentation quality, and boundary visibility, especially in the case of such complicated tumor structures.

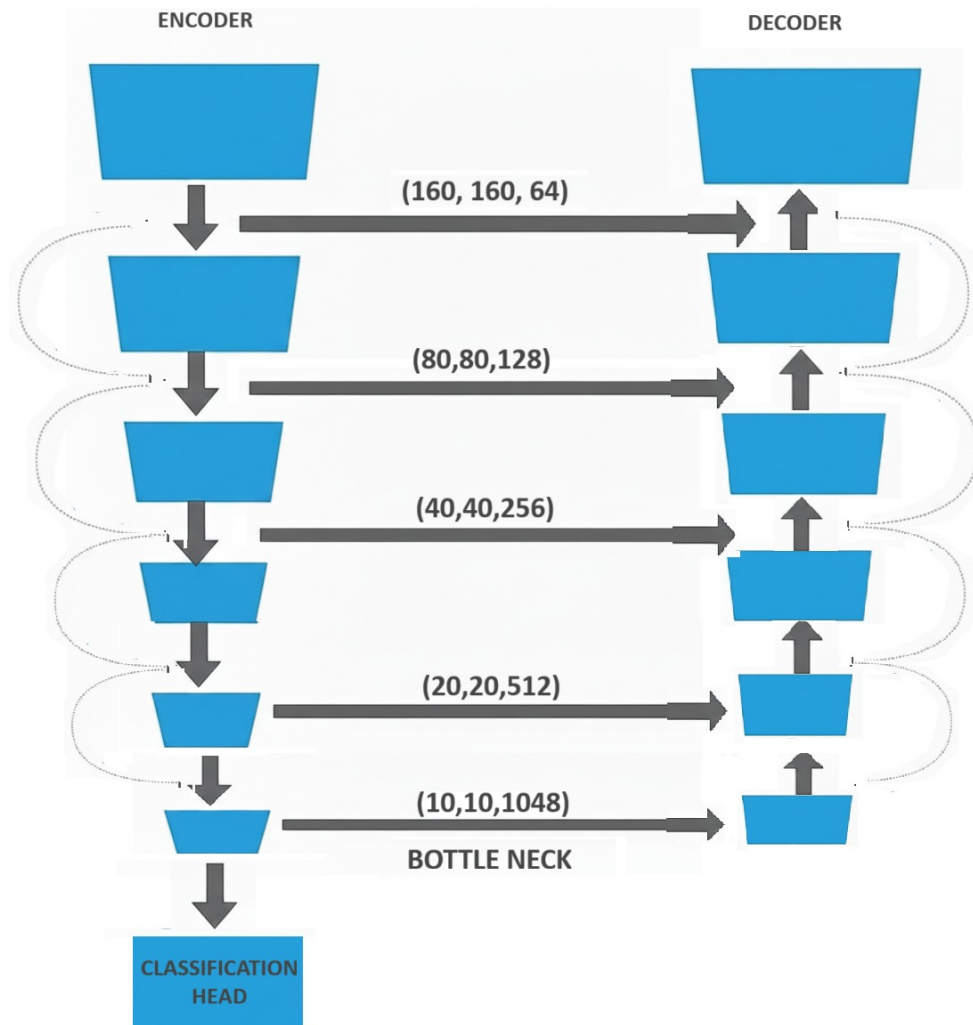


Figure 6.3: Architecture of ResU-Net

- **Attention Residual U-Net:** The third and more developed one injects attention gates (AGs) into the Residual U-Net framework. Those gates are fitted in the skip connections and learn to emphasize useful spatial features and blocking irrelevant or noisy activations. This leads to the decoder obtaining the most informative features and this will greatly improve the precision of the segmentation particularly in low contrast or noisy areas. More focused representations are also beneficial to the classification head which further improves predictive accuracy. The architecture has been proven to provide high-performance in all the measures that have been assessed and is most appropriate where high accuracy is essential such as in case of clinical applications.

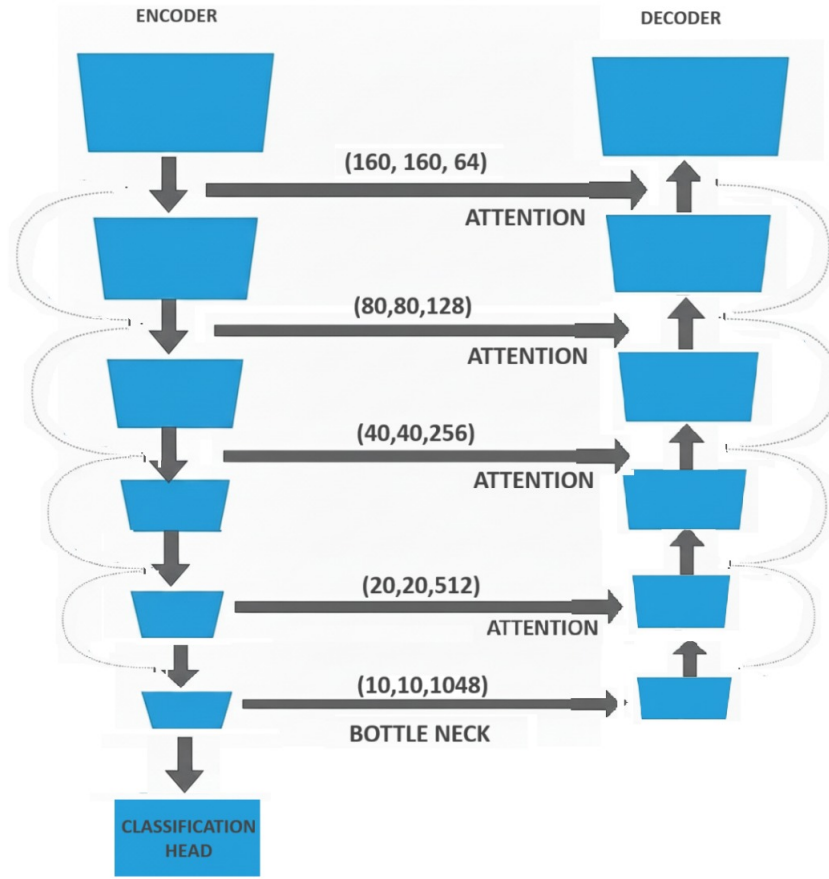


Figure 6.4: Architecture of Attention Residual U-Net

6.4 Dual-Head Output Strategy

Each of the three architectural variants makes use of a common encoder and then two output heads:

- **Segmentation Head:** This head is charged with the responsibility of producing a binary mask that displays the tumor regions at the pixel level. It is made up of transposed convolutional layers and skip connections between the encoder and the spatial resolution. The final layer is a sigmoid activation function that produces the pixel-wise probabilities which is used to binary classify the pixels as either tumor or background.
- **Classification Head:** The classification head works based on the most deep features of the encoder, and the head is meant to extract the global contextual information. It is normally composed of global average pooling, then complete connected dense layers together with a sigmoid output unit. This branch produces a binary prediction of whether there is a tumor in the input image or not. Training the two tasks simultaneously motivates the model to learn representations that are accurate spatially and rich semantically, and enhances the accuracy of the two outputs.

This dual-task formulation has some benefits: it normalizes the encoder, improves the learning of

tumor-related patterns, and gives auxiliary supervision that increases the capacity of the model to generalize to different patient cases and imaging orientations.

6.5 Training Configuration and Hyperparameters

All the three versions of U-Net were trained on the same dataset, preprocessing pipeline, and hyperparameters to guarantee consistency in assessment and consistency in comparison. Images were scaled to 160 160 pixels, and intensity normalized and increased by random flips, rotations and small changes to intensity to enhance generalization.

The training of the segmentation task was done on a composite loss that was based on Dice Loss and Binary Cross-Entropy (BCE), which reduces imbalance of classes and encourages proper boundary prediction. Optimization was done on the classification head with standard BCE loss. The combined loss was calculated as a weighted average of the two losses, the weight normally being equal, unless otherwise given.

The Adam optimizer was used to train models with an empirically selected learning rate, and a learning rate scheduler was used to decrease the learning rate when the validation loss leveled off. Experiments were held constant in batch size, number of epochs and optimizer parameters. Termination of training early was done when the validation performance stopped improving and helped prevent overfitting as well as saving computation time. All the training metrics like loss curves, accuracy scores and segmentation evaluations were recorded to be analyzed and compared in terms of performance.

6.6 Comparison Between the Three Variants

A thorough comparative study was done to determine the effectiveness of the three U-Net variants. The Base U-Net found a strong foundation and segmented successfully clear and high-contrast image. Nevertheless, it had problems of ambiguous boundaries and complex tumor shapes with lower convergence and moderate accuracy of classification.

The Residual U-Net was much better on these fronts. The residual blocks increased feature extraction and feature stability throughout the training. It was always able to generate better segmentation masks and show better classification performance because of the strength of the encoder.

The Attention Residual U-Net was better than the other two models. It could highlight the salient features by incorporating attention gates and dampen irrelevant regions, the result of which was better edges detecting and a focus on the category, and this was superior to edge detection with other algorithms. This model was superior in images where structures overlap, where the background is noisy, or the size of the tumor is small, and so it is the best candidate that can be used to support clinical-grade segmentation and diagnosis.

The attention-based model scored the highest in all measurements of the evaluation metrics, including Dice coefficient, pixel accuracy, and classification AUC, which proves the usefulness of the residual learning and spatial attention combination in the U-Net architecture.

Chapter 7

RESULTS AND ANALYSIS

7.1 Quantitative Performance Comparison

The three U-Net variants, Base U-Net, Residual U-Net and Attention Residual U-Net, were evaluated based on a set of evaluation metrics.

- Dice Score: The dice score measures the overlap between the predicted and actual segmentation mask.
- IoU (Jaccard Index): Measures the segmented area of intersection and union.
- Precision, Recall and F1-Score: Assessed classification.
- Classification Accuracy: Refers to the correct image-level tumor classification.

From the table and visualizations, the results are summarized below:

Metric / Model	Attention ResU-Net	ResU-Net	Base U-Net
Dice Score	0.8664	0.8648	0.8645
IoU (Jaccard Index)	0.7666	0.7636	0.7633
Precision	0.9812	0.9521	0.8711
Recall	0.9332	0.8854	0.8537
F1-Score	0.9244	0.8975	0.8823
Classification Accuracy	0.9262	0.9875	0.9937

Table 7.1: Performance Comparison of U-Net Variants on Segmentation and Classification Tasks

The attention-based model scored the most points in segmentation specific measures (Dice, IoU, F1, Recall, Precision) suggesting that the model was well suited in the process of identifying tumor boundaries and reducing false positives. The plain U-Net was a little more successful in classification but at the expense of accuracy and edge definition.

7.2 Result Analysis – Base U-Net

Learning Curves: Training and validation loss rate diminished constantly, which was good convergence. Dice and IoU were getting better with every epoch, which proves that the quality of segmentation was getting better.

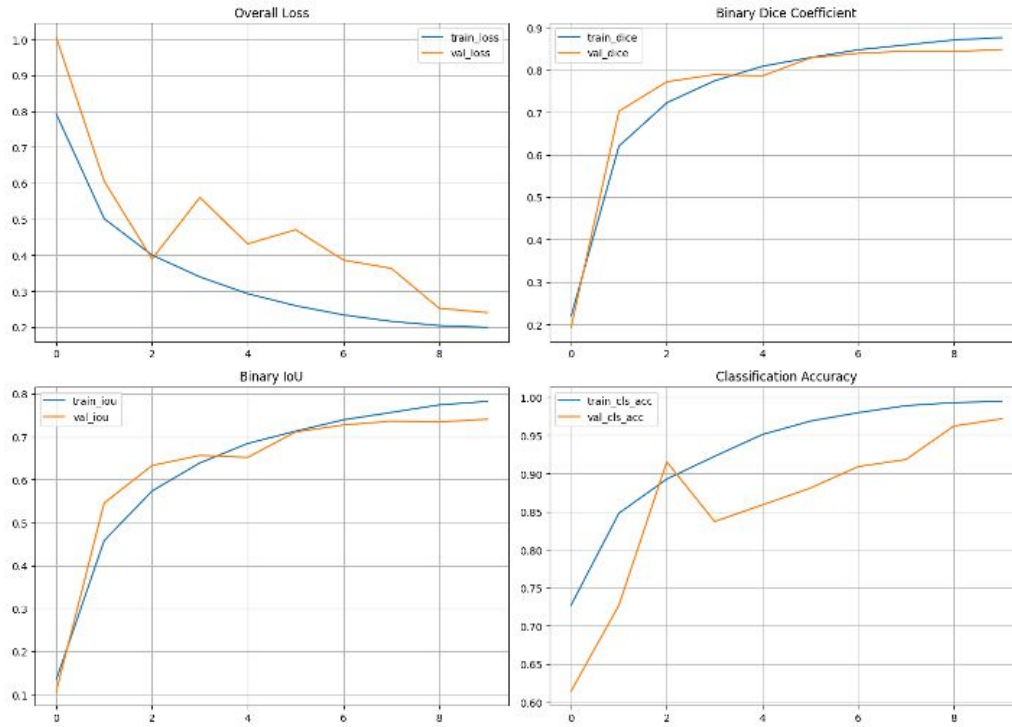


Figure 7.1: Loss and accuracy graphs (Base U-Net)

Confusion Matrix: The classification accuracy (0.9937) was very high with no misclassification and all the types of tumors were correctly identified.

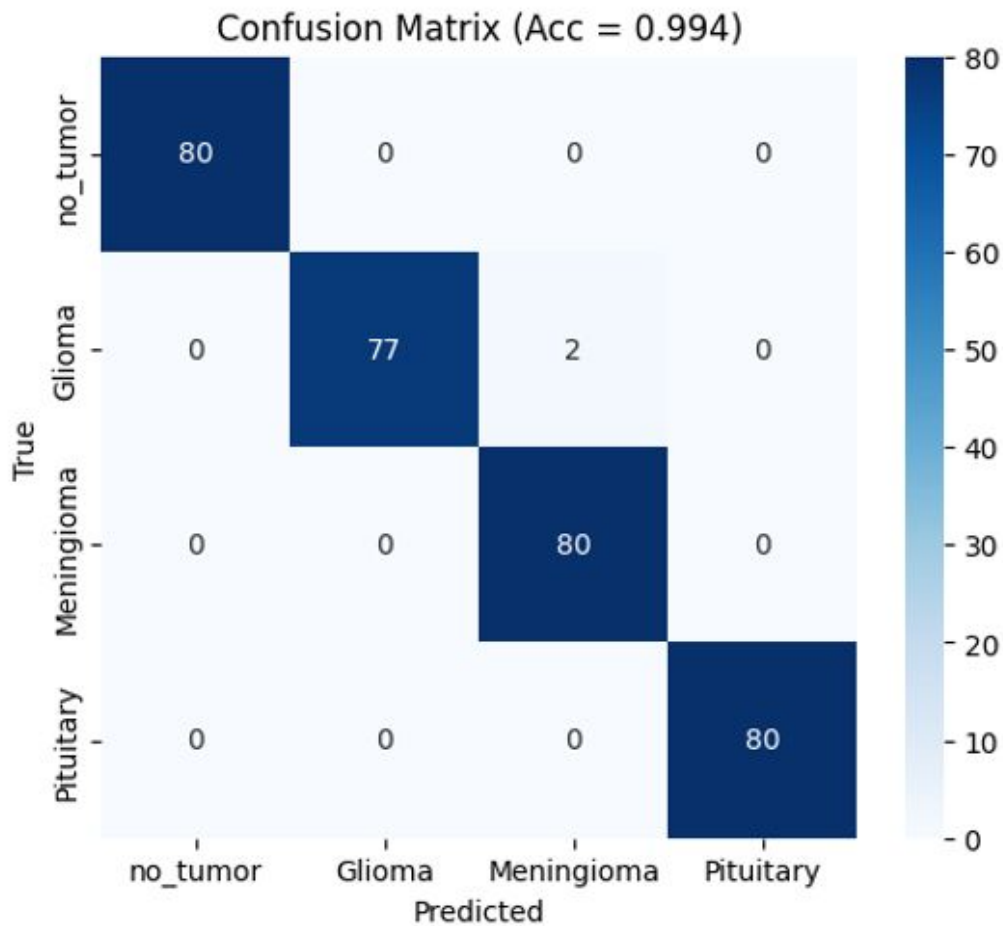


Figure 7.2: Confusion matrix for Base U-Net

Prediction Sample: The mask output of Pituitary tumor was consistent with the ground truth and there was good localization of spatial prediction.

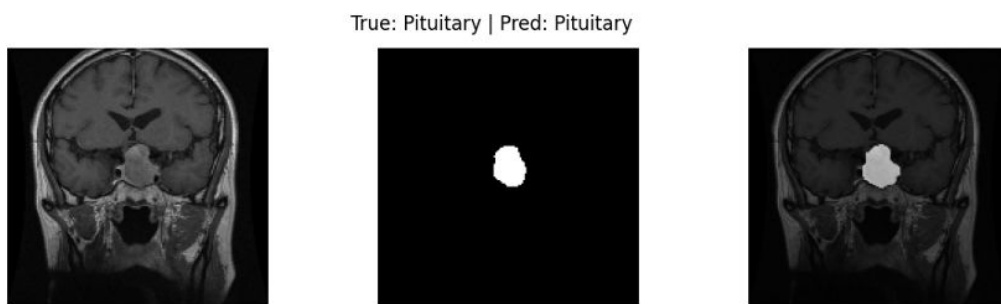


Figure 7.3: MRI + predicted mask visualization

Nevertheless, it was limited in the expression of features because it could not deal with complex boundaries and smaller tumor extents using this model.

7.3 Result Analysis – Residual U-Net

Learning Curves: Rapid convergence and more continuous improvement in Dice and IoU metrics than the base model.

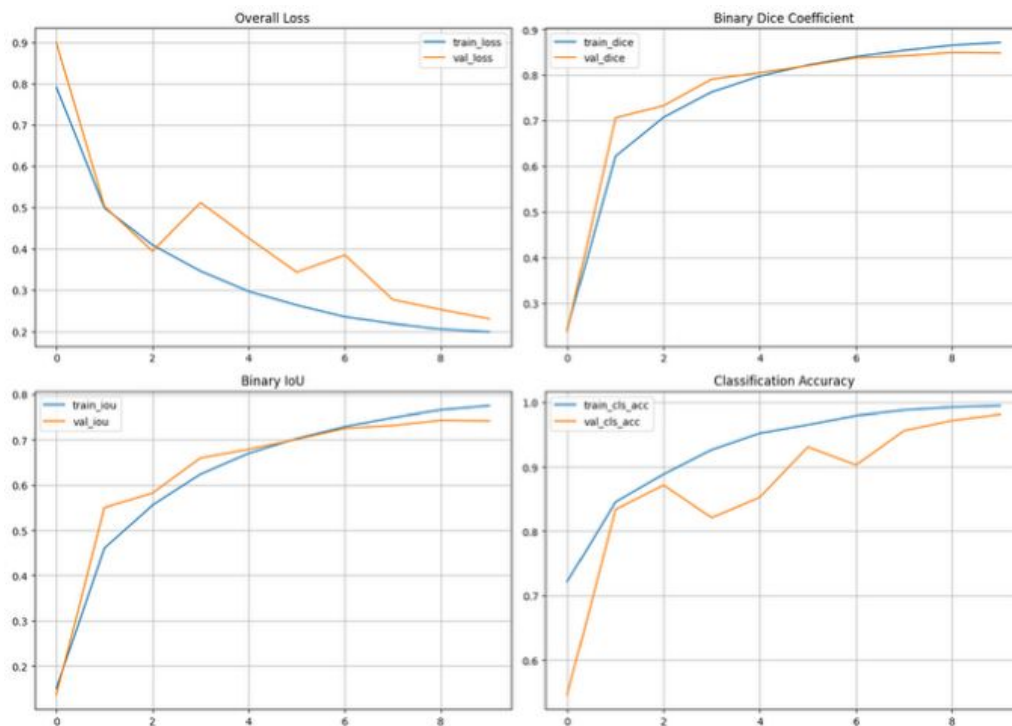


Figure 7.4: Loss and accuracy graphs (ResU-Net)

Confusion Matrix: reached an accuracy of 0.987, mistaken 2 cases of Meningioma as Pituitary, there was some slight confusion between morphologically similar types.

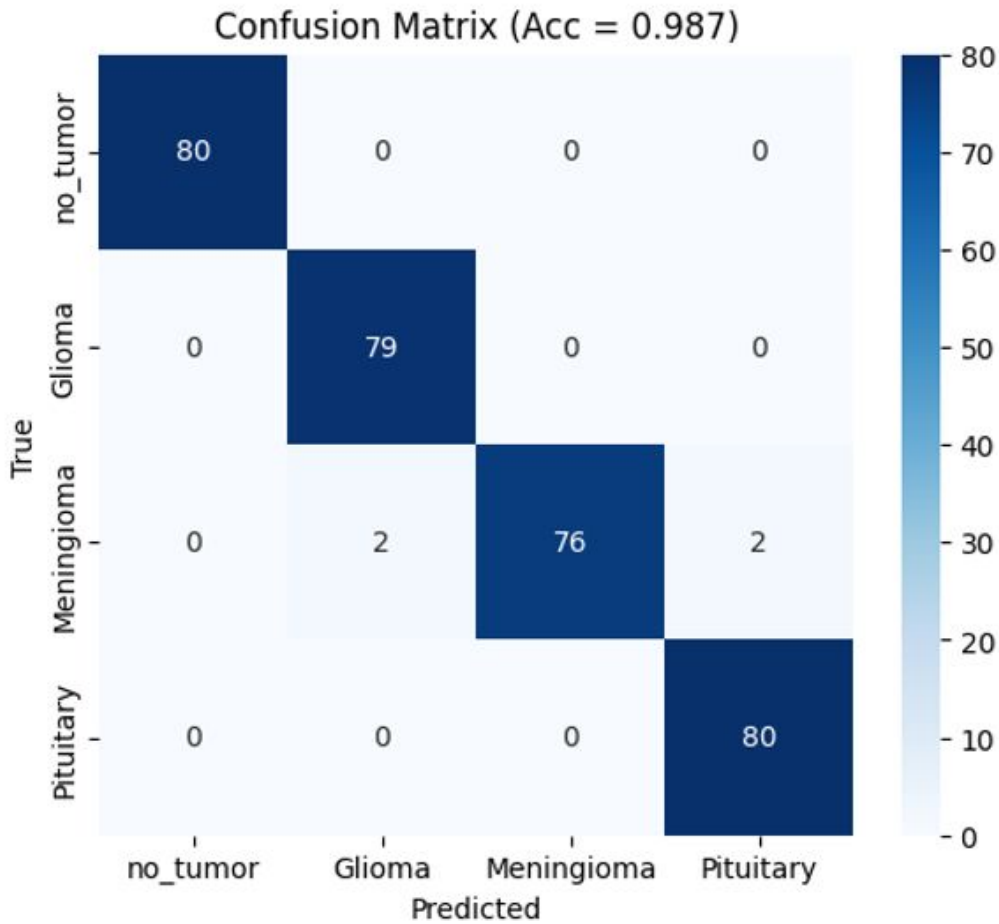


Figure 7.5: Confusion matrix for ResU-Net

Prediction Sample: Provided more accurately structured cleaner boundary predictions.

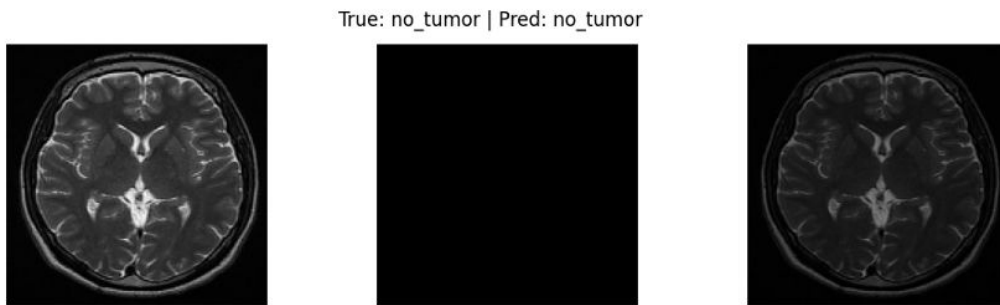


Figure 7.6: MRI + predicted mask visualization

This model showed better segmentation and recall than the base U-Net with more in-depth propagation of features.

7.4 Result Analysis – Attention Residual U-Net

Learning Curves: During training, the loss reduced gradually, whereas during validation, it fluctuated with some irregularities, and probably because of the complexity of attention processing. Dice and

IoU metrics gave the best results.

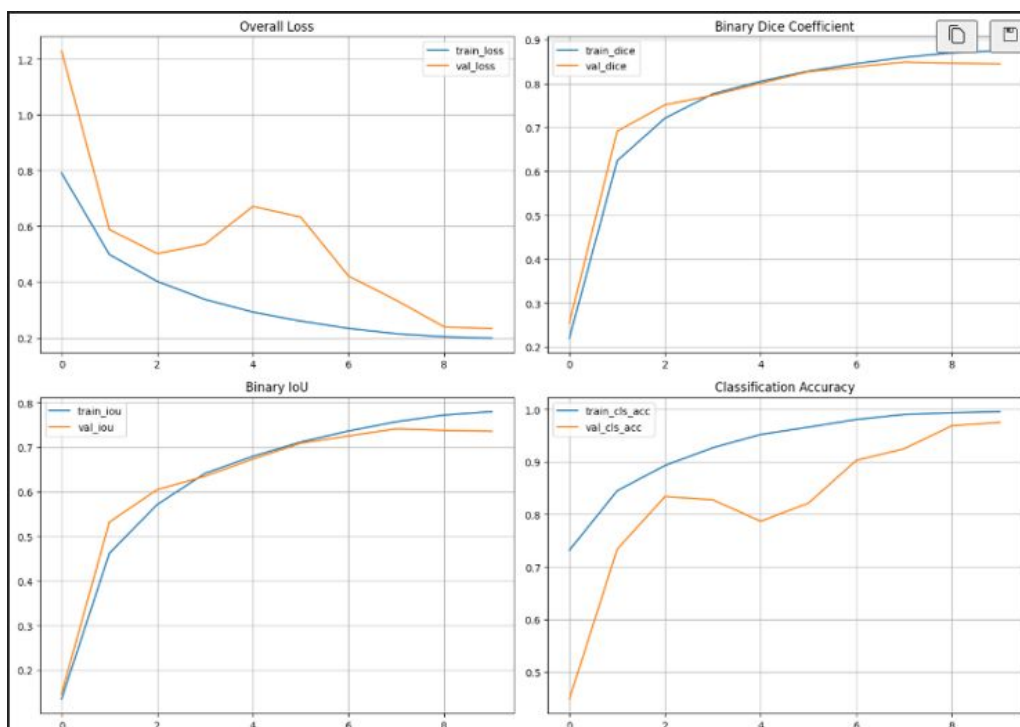


Figure 7.7: Loss and accuracy graphs (Attention ResU-Net)

Confusion Matrix: There was a slight decrease in accuracy to 0.922 with the majority of confusion in the Meningioma category, which tends to be structurally similar to other tumors.

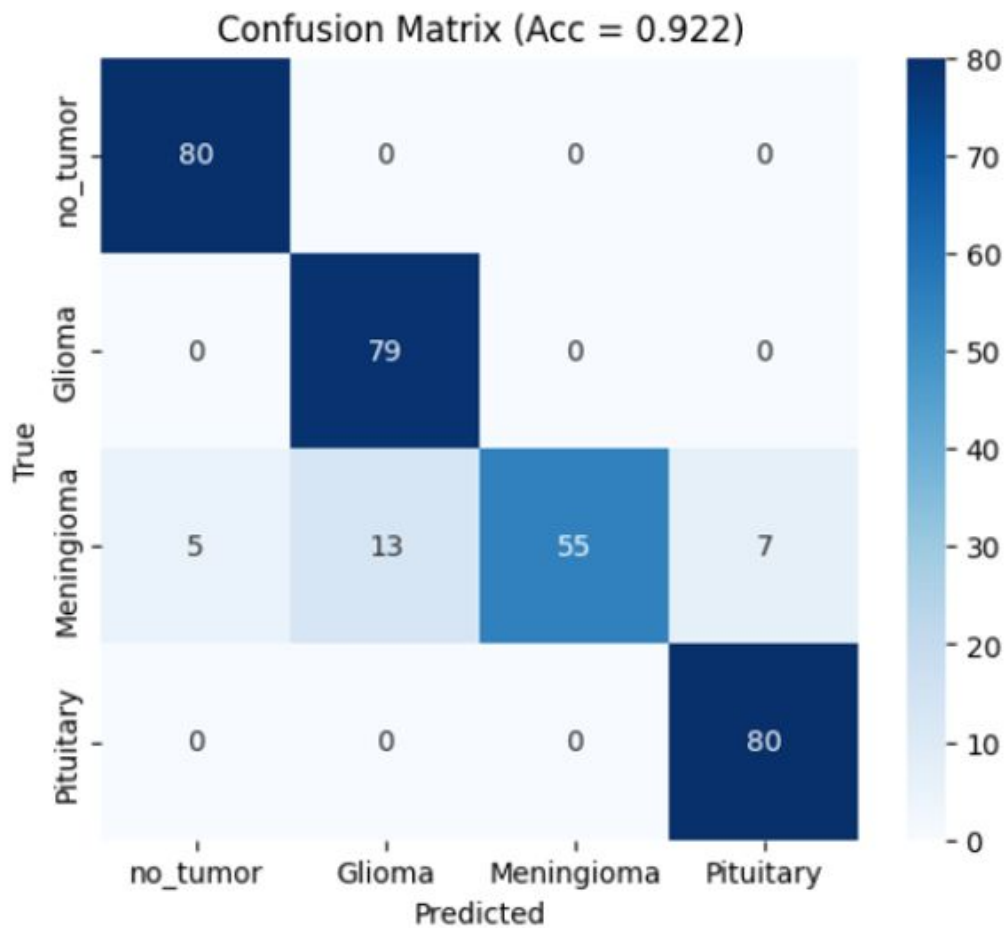


Figure 7.8: Confusion matrix for Attention ResU-Net

Prediction Sample: Segmentation output was the best with sharp-cutting masks even though there was slight confusion in classification.



Figure 7.9: MRI + predicted mask visualization

This model has demonstrated a good ability to learn to emphasize pertinent areas in space, and regress irrelevant features and noise, which is why it is the best model to use in strong segmentation in diverse conditions.

7.5 Final Comparison and Model Selection

Feature	Base U-Net	Residual U-Net	Attention ResU-Net
Segmentation Clarity	Moderate	Improved	Best
Boundary Accuracy	Weak on complex	Stronger	Strongest
Classification Accuracy	Highest (0.9937)	High (0.9875)	Good (0.9262)
Dice / IoU / F1-Score	Moderate	High	Highest
Attention Mechanism	No	No	Yes
Robustness to Noise	Low	Medium	High
Best For	Simpler tumors	Balanced cases	Clinical precision

Table 7.2: Evaluation Metrics Summary for Base U-Net, ResU-Net, and Attention ResU-Net

Attention Residual U-Net is the best model to use when it comes to clinical application and high-precision segmentation. Even though its classification performance is a little bit lower, it outperforms all the vital segmentation metrics, namely, Dice, IoU, Precision, and F1, which makes it the most suitable option in the situation when an accurate delineation of tumor boundaries is essential to diagnosis or surgical planning. Attention gates enable it to be more robust to noise and imaging variability, which confirms its high-quality analysis of medical images in real-world scenarios.

Chapter 8

CONCLUSION AND FUTURE ENHANCEMENT

8.1 Conclusion

We developed a multi-task deep learning architecture and tested it on a brain tumor segmentation and classification task on MRI scans. The main task was to be able to localize the tumor areas properly on a pixel-to-pixel basis and at the same time to be able to forecast the type of tumor (or the absence thereof) on an image-to-image basis. The developed architecture was developed upon the popular U-Net framework, and supplemented with a residual learning and attention systems, as well as dual output heads to obtain both segmentation and classification.

It implemented and compared three model variants, i.e. the standard Base U-Net, the Residual U-Net (ResU-Net), and the Attention Residual U-Net. The same training pipeline was used on all the models based on a curated MRI dataset that consisted of four different tumor classes: glioma, meningioma, pituitary tumor, and no tumor. The quantitative analysis using the major indices like Dice Score, IoU, Precision, Recall, F1-Score, and Classification Accuracy proved that the Attention Residual U-Net was always superior to the other systems. Its integration of attention gates allowed the model to be more attentive to the differences in relevant spatial regions especially improving segmentation results in the presence of noise or low contrast MRI slices.

The reliability and the strength of the suggested design of the two-headed one was further confirmed by the training curves, confusion matrices, and segmentation overlays. The model learned rich features, both spatially and semantically, by jointly optimizing segmentation and classification which increased generalization as well as boundary accuracy. The encoder was also checked by the integration of classification as an auxiliary task that made the model overall more stable and efficient to learn.

To conclude, the Attention Residual U-Net stands out as the most effective architecture of the three as it offers the best diagnostic support with accuracy in segmentation, and high-quality tumor classification.

8.2 Future Enhancement

The obtained results are encouraging, but a number of directions may be followed to improve the efficacy, scalability, and clinical applicability of the model:

- **3D MRI Volume Integration:** Existing models are based on 2D slices, which do not necessarily model continuity of tumors on a volumetric basis. The architecture can be expanded to support entire 3D volumes to give more detailed spatial information, enhancing segmentation sensitivity and tumor morphology knowledge.
- **Class Imbalance Mitigation:** The classification and segmentation performance might be further enhanced through the incorporation of advanced data balancing methods or adaptive loss functions (e.g., focal loss or class weighted Dice) particularly in the case of the minority tumor types (such as meningioma).
- **Generalization of Multi-Center Data set:** In order to assure the model clinical applicability, it has to be trained and tested on multi-institutional datasets of varying acquisition protocols. Domain shift could be addressed with domain adaptation and normalization algorithms (e.g., histogram matching or CycleGANs).
- **Clinical Metadata Incorporation:** Contextual cues that have been added to the model as auxiliary inputs like age, symptoms, or previous diagnosis can add to classification and increase model interpretability.
- **Real-Time implementation and optimization:** Pruning, quantization, and knowledge distillation are examples of model compression that might be used to implement lightweight implementations of the model to be used in real-time in a clinical environment or on the edge.
- **Explainability and Interpretability Tools:** Explainable AI tools, such as Grad-CAM, attention visualizations or SHAP, might also be added to its integration to make the model more transparent, and the predictions easier to interpret by radiologists and clinicians.
- **Post-Processing Morphological Filtering:** False positives or disconnected regions in the segmentation masks can be solved with minor false positives by applying post-processing tools in morphological operations or conditional random fields (CRFs).
- **Efficiency of Annotation through Active Learning:** Active learning frameworks can help to decrease labeling effort by choosing the most informative samples to be annotated so that a large dataset can be grown with only a small number of expert interventions.

The model can be developed into a fully-fledged and reliable diagnostic instrument by seeking these improvements to fit into clinical decision-support systems.

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